

Vig Removal Changes the Answer

Abstract · MIT Sloan Sports Analytics Conference 2027 Research Paper Competition · Basketball

Introduction

Almost every test of betting-market bias starts with the same step: taking the sportsbook's cut out of the price. There isn't one correct way to do that, and most papers pick a method in one sentence and move on. I wanted to know how much the answer depends on that sentence. It matters because de-vigged odds get treated as true probabilities almost everywhere, and studies still disagree on which way favorite bias points.

Methods

I tested one question, fixed before I looked at any outcomes: do NBA favorites with a vig-free win probability above .70 win as often as their prices say? The data are consensus moneylines for 15,351 regular-season games, 2007-08 to March 2020, from the Sportsbook Reviews Online archive. Every price and result matched a separate extraction of the same archive (30,978 prices, 15,490 outcomes).

Each game has its own probability, so if the prices are right, favorite wins follow a Poisson-binomial distribution, which I computed exactly with a discrete Fourier transform. At the break-even margin this test has over 99.9 percent power. A 74-game bucket, the kind past studies often report, has about 11 percent (Figure 2). I also ran an equivalence test against the break-even margin and reran the main test under five vig-removal rules on the same games, so only the rule changes.

Results

Under the proportional rule set in advance, favorites above .70 (6,840 games) won 80.83 percent of the time against an implied 80.26 percent. That's a gap of +0.58 percentage points ($z = 1.22$, $p = .223$, 95 percent CI -0.35 to +1.51). A bettor needed 2.99 points to break even, and the equivalence test rules out a gap that big ($p = 4.7e-08$). This holds with errors clustered by season and by team.

Then I changed only the vig-removal rule. On the same games the gap moves from -1.21 to +0.58 points (Figure 1). That's not a rounding issue, it's a sign flip. Power normalization makes the gap significant ($p = .008$), though it only matches the break-even point. Skipping vig removal shows an apparent favorite bias of -2.43 points ($p < .001$) that's really just the sportsbook's cut.

Conclusion

Betting favorites or underdogs in these prices gave no edge. The bigger result is that the evidence for favorite bias depends on how the vig is removed, and a single-rule bucket test can't see that. That matters to modelers, sportsbooks, and especially researchers testing for bias. The fix is cheap. Report results under several rules, not one. Reporting how the answer moves isn't a caveat on the result. It is the result.

Data and code: <https://github.com/jacobburdier05/nba-moneyline-calibration>. One command rebuilds every number from the raw archive in about six minutes.

Figure 1

Same 6,840 games. Change only the vig rule and the sign flips.

Favorites above .70 vig-free probability, NBA regular season, 2007-08 to March 2020

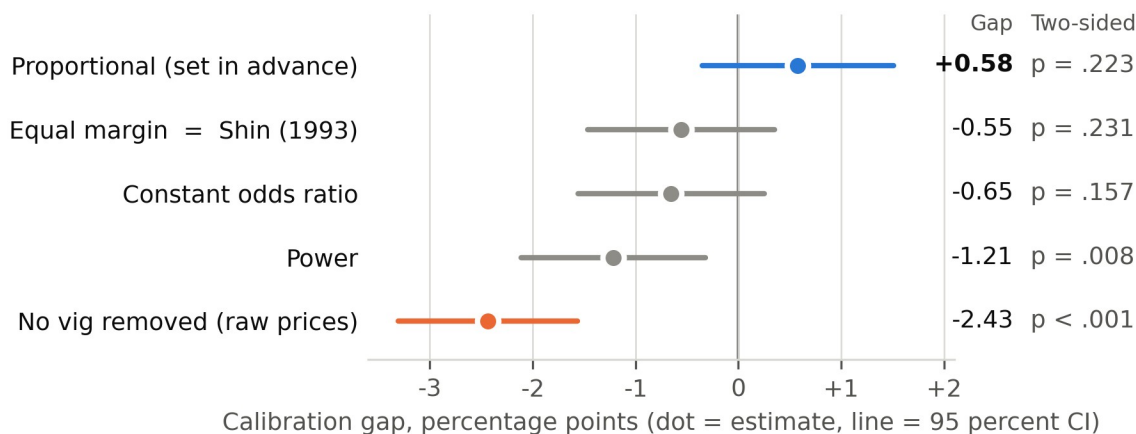


Figure 2

Same question, very different chances of seeing an answer.

Power of a two-sided test at alpha = .05

